

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/I_DEB_SAAWKGe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 6333, C2 = 3665, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.01,      0.8);
Distrib (pAm,         TruncNormal_cv,       100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,       0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,       0.18505,  0.1, 0.001,  2);
Distrib (kap,         Uniform,               0,        1);
Distrib (kapA,        Uniform,               0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,       100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,       30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001, 0.3);
Distrib (K,           Uniform,               0.00001, 2);
Distrib (dv,          TruncNormal_cv,       2,        0.5, 0,      10);
Distrib (wE,          LogUniform,           0.000000001, 0.1);
Distrib (gOM,         LogUniform,           0.0001, 1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,      Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.46704000	153.8830	0.88771800	0.1716960	0.96032900	0.75502700
2.5%	0.39824225	90.3035	0.73356100	0.1448122	0.93542030	0.60658900
97.5%	0.68661000	158.6730	1.01800500	0.2209305	0.97942100	0.84889000
Min	0.03146980	70.2446	0.64931900	0.1213700	0.07193230	0.41118600
Max	0.79532500	172.4890	1.12862000	0.2417600	0.98375900	0.88695300
Result_mean	0.52432488	127.9790	0.88304787	0.1861100	0.96080074	0.73805096
Result_sd	0.07607911	18.2719	0.07274134	0.0203216	0.01365582	0.05995473

	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.
Result_mode	92.97780	26.844600	0.004428910	0.3863130	1.8084600
2.5%	77.71887	19.728100	0.001820930	0.3124170	0.7631316
97.5%	150.17305	41.408800	0.007062642	0.6746050	2.2981400
Min	57.28450	14.756600	0.000595130	0.2212010	0.6213240
Max	179.51700	56.362400	0.239108000	0.7169130	3.7870300
Result_mean	113.97237	29.522752	0.004266491	0.4935277	1.4959720
Result_sd	18.81217	5.853316	0.001947803	0.1041328	0.4397254

	wE.1.	gOM.1.	Sigma_W.1.
Result_mode	1.029430e-05	0.000104871	0.10578700
2.5%	9.236720e-06	0.000118129	0.09522530
97.5%	2.074919e-05	0.139198000	0.12114100
Min	2.050180e-09	0.000100630	0.08494590
Max	2.331110e-05	0.188617000	6.03405000
Result_mean	1.316394e-05	0.028849037	0.10775549
Result_sd	2.651380e-06	0.041515563	0.05460272

Number of observations, n = 277

Number of parameters, k = 13

Log-likelihood, LL = 8.839794

BIC = 55.43264

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	2.49	4.68
E_coc.1.	1.06	1.17
Ehp.1.	1.06	1.18
Fm.1.	2.08	3.63
gOM.1.	2.06	5.91
K.1.	2.57	4.55

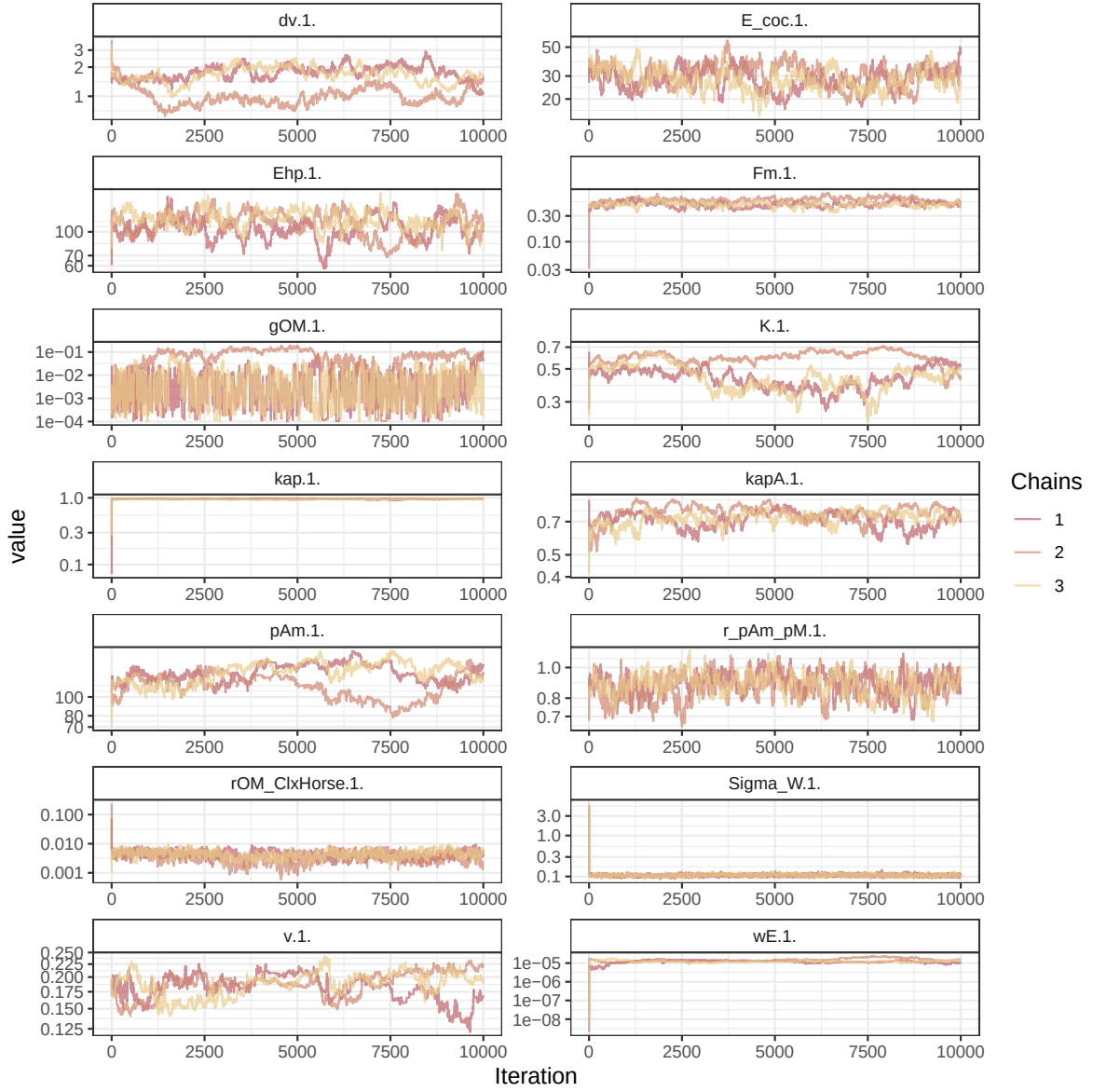


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

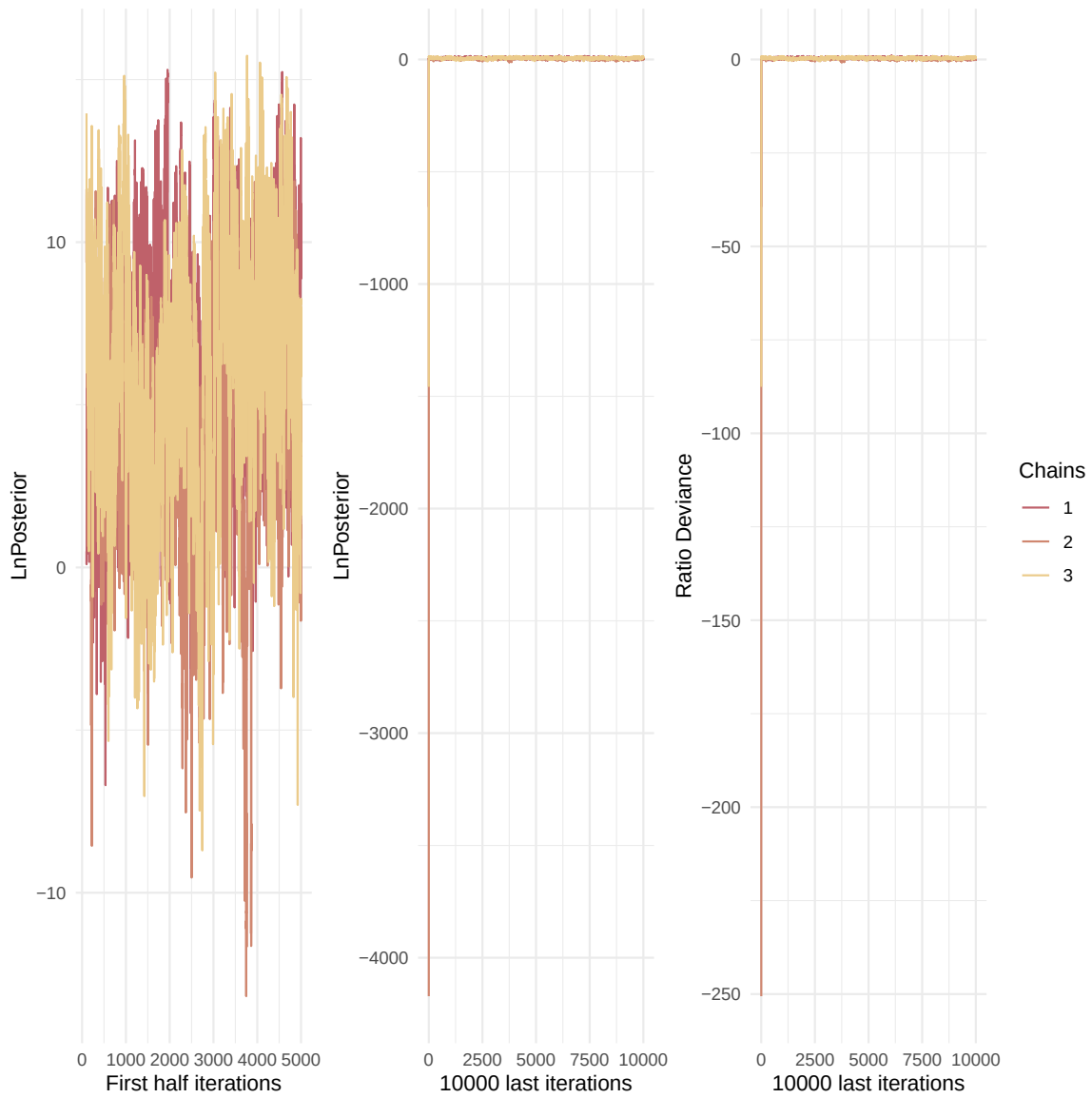


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kap.1.	1.79	3.18
kapA.1.	1.16	1.47
pAm.1.	2.21	3.87
r_pAm_pM.1.	1.01	1.04
rOM_ClxHorse.1.	1.06	1.19
Sigma_W.1.	1.06	1.18
v.1.	1.40	2.09
wE.1.	2.66	7.22

Multivariate psrf

3.19

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

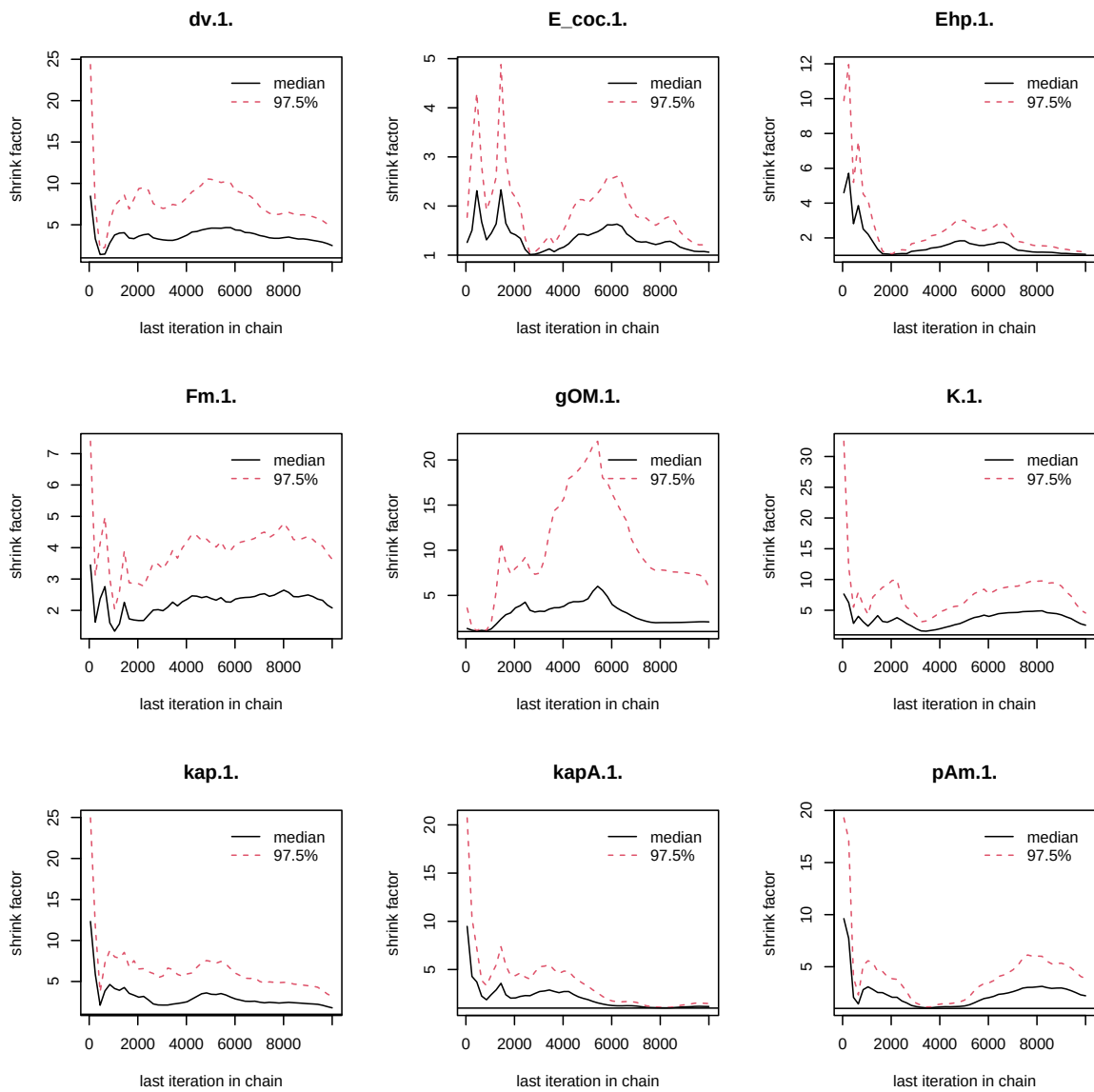


Figure 3: Chains convergence.

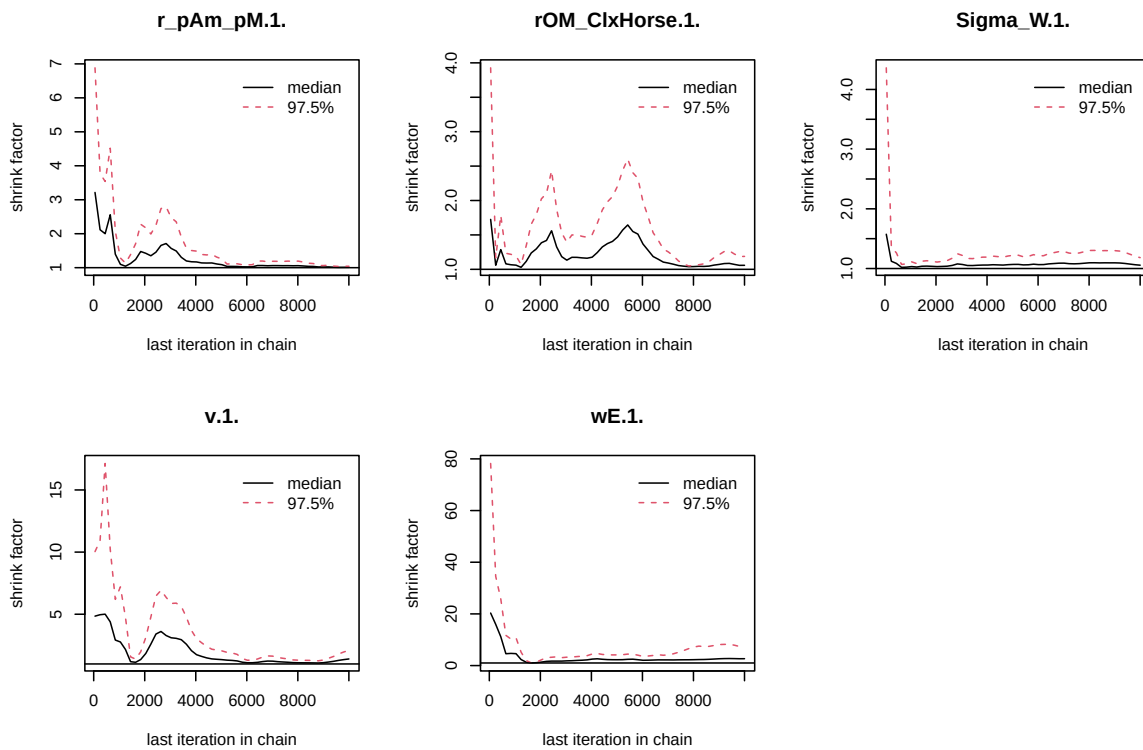


Figure 4: Chains convergence.

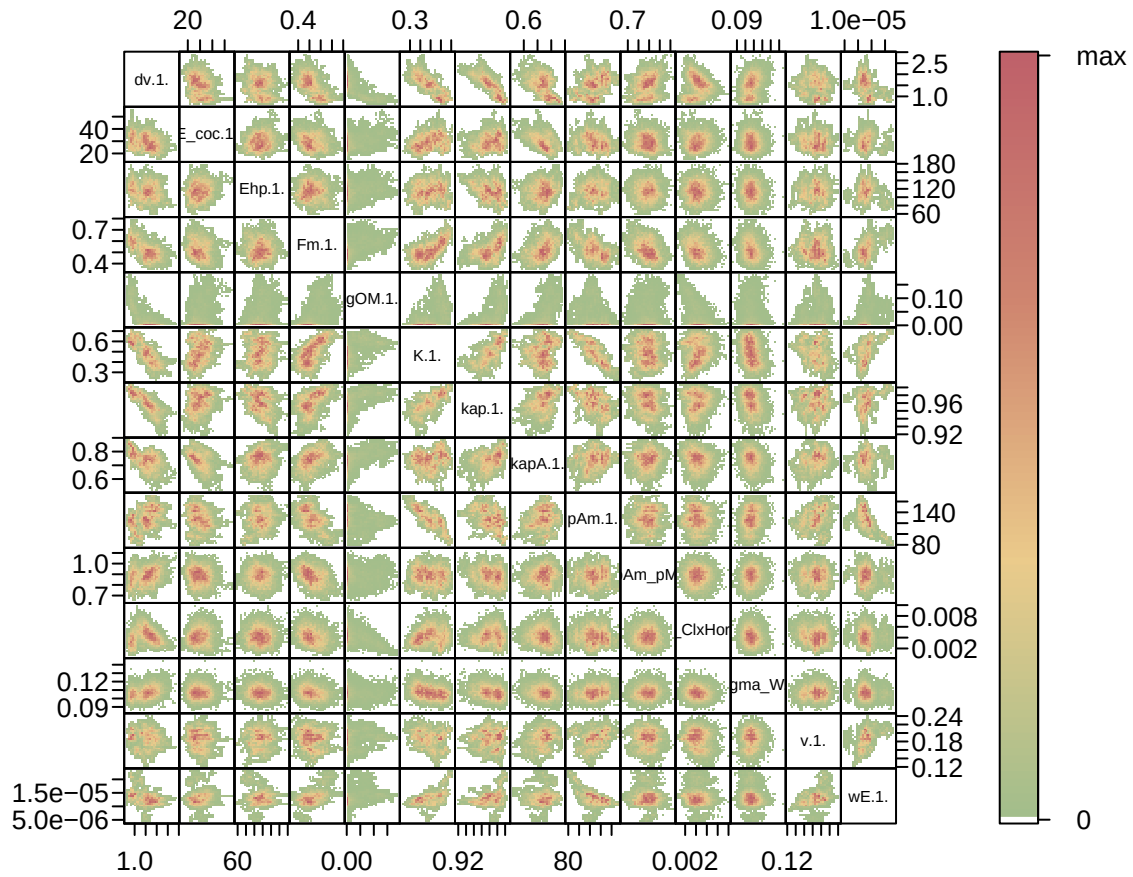


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

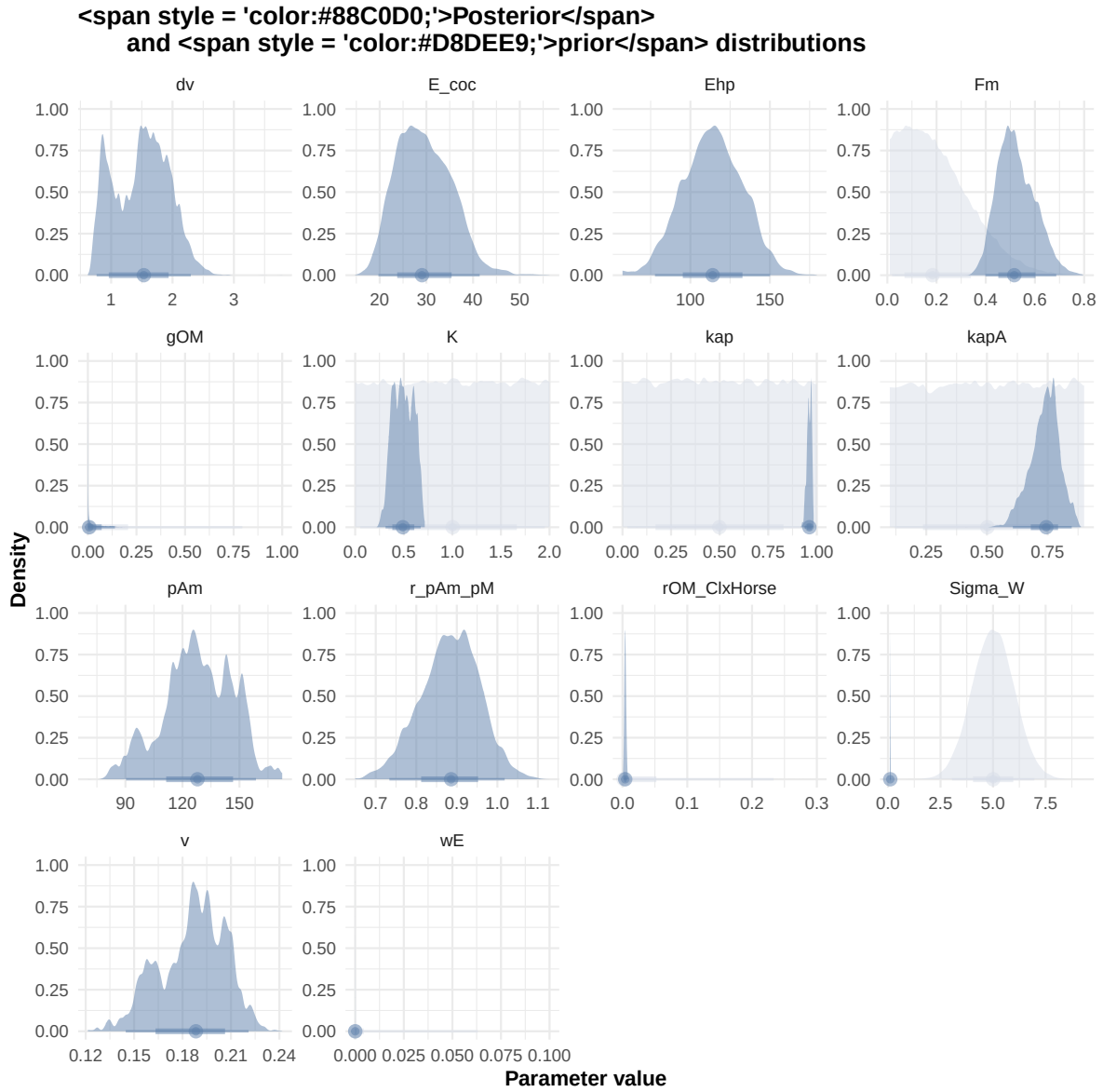


Figure 6: Distributions of the priors and the posteriors.

```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.\\.|\\.\\.\\.1\\.\\.\\.\"", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),

```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
  v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
  #####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
 i Row 1 of 'x' matches multiple rows in 'y'.
 i Row 1 of 'y' matches multiple rows in 'x'.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

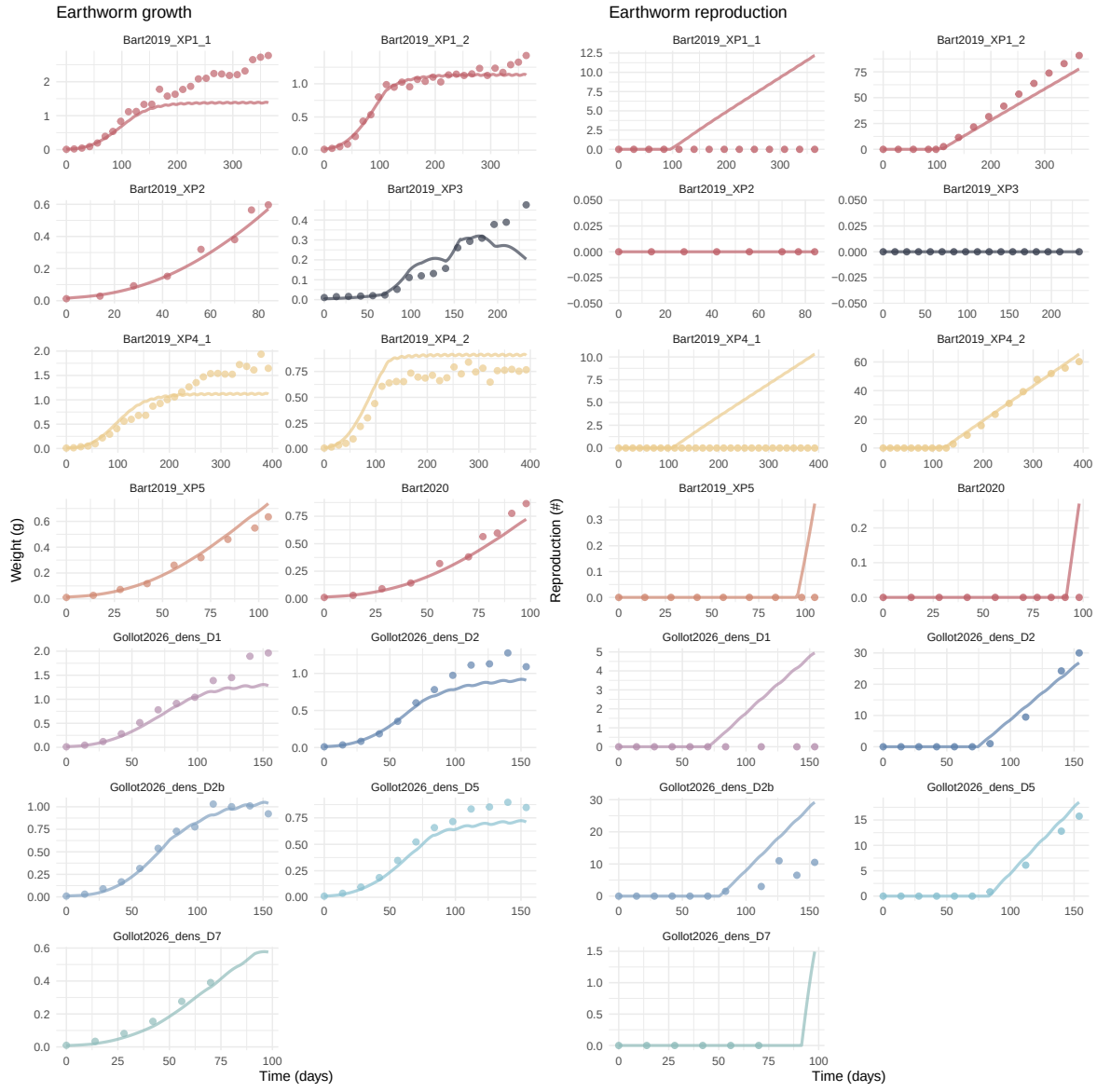
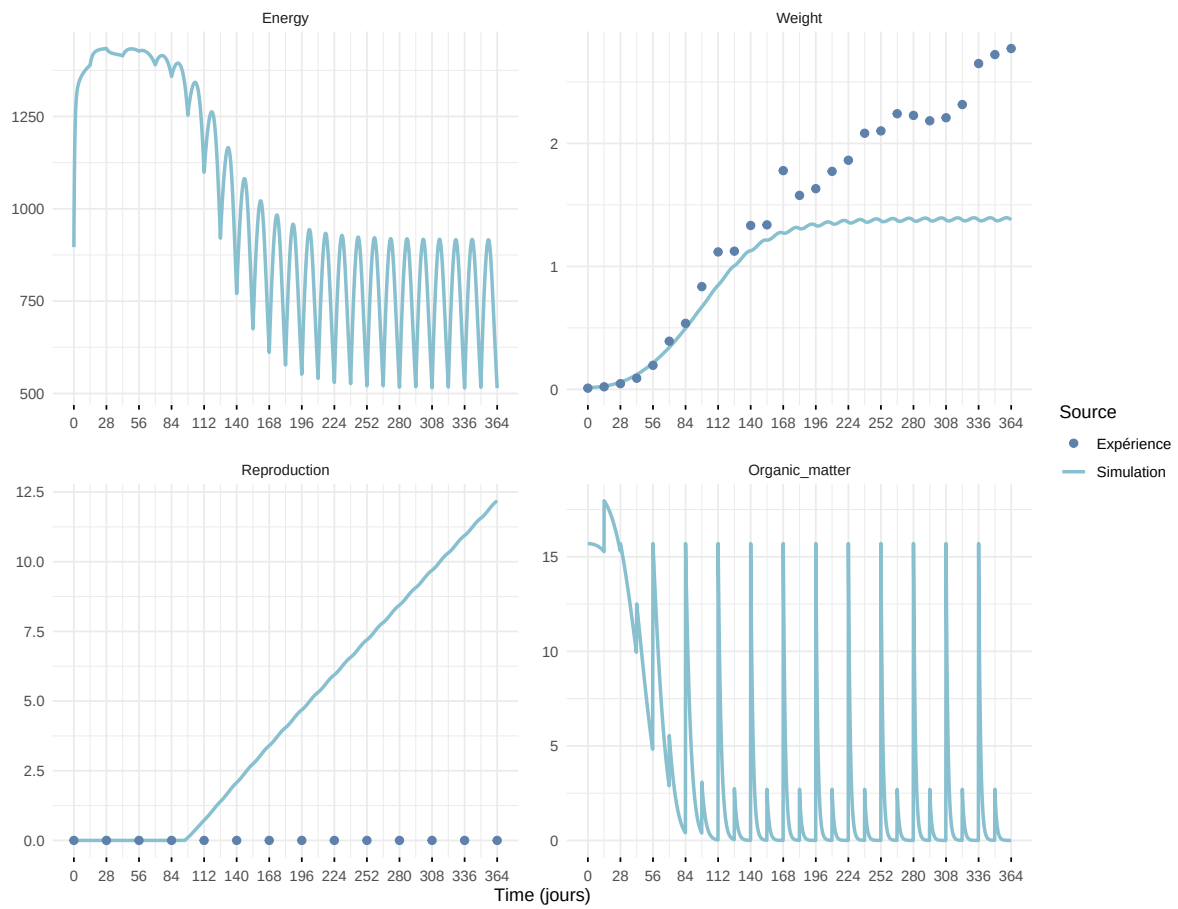
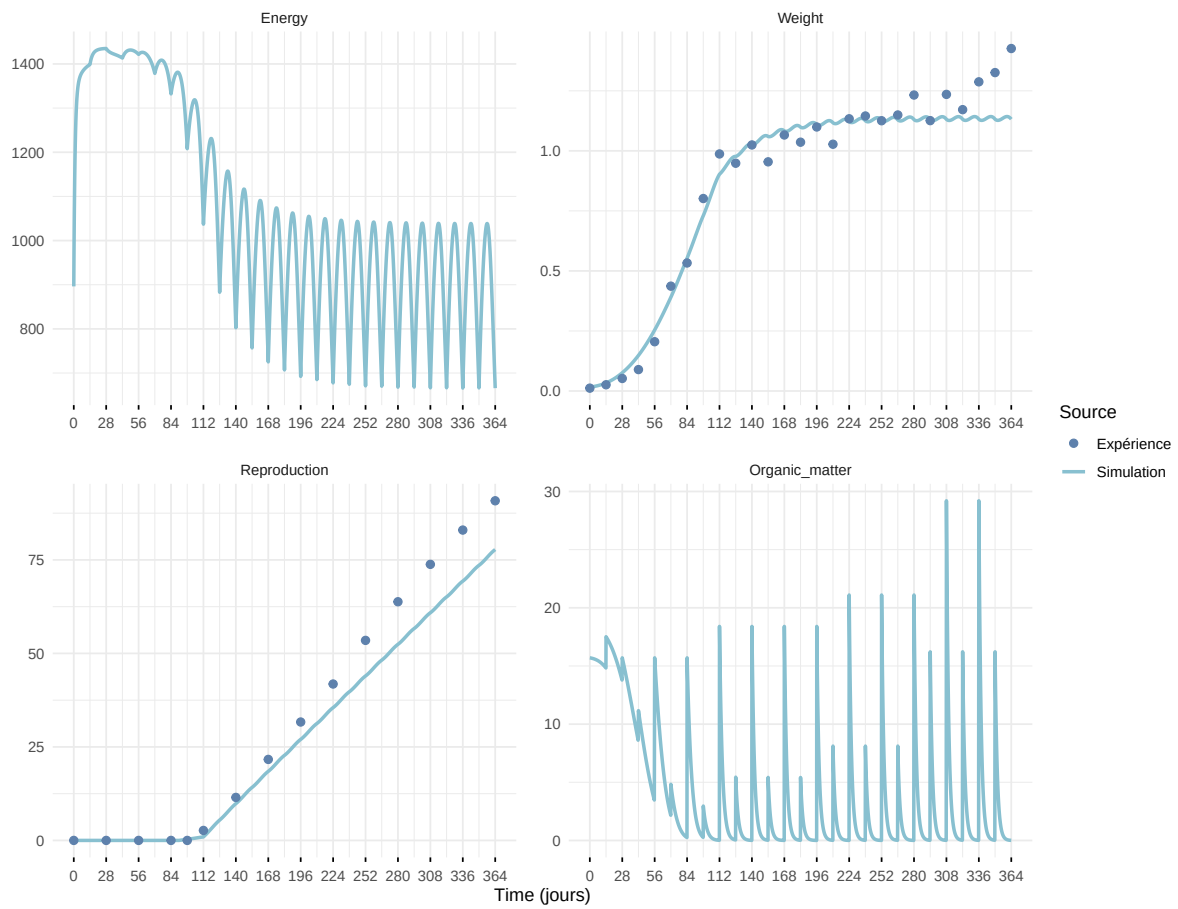


Figure 7: Simulations Weight and Reproduction.

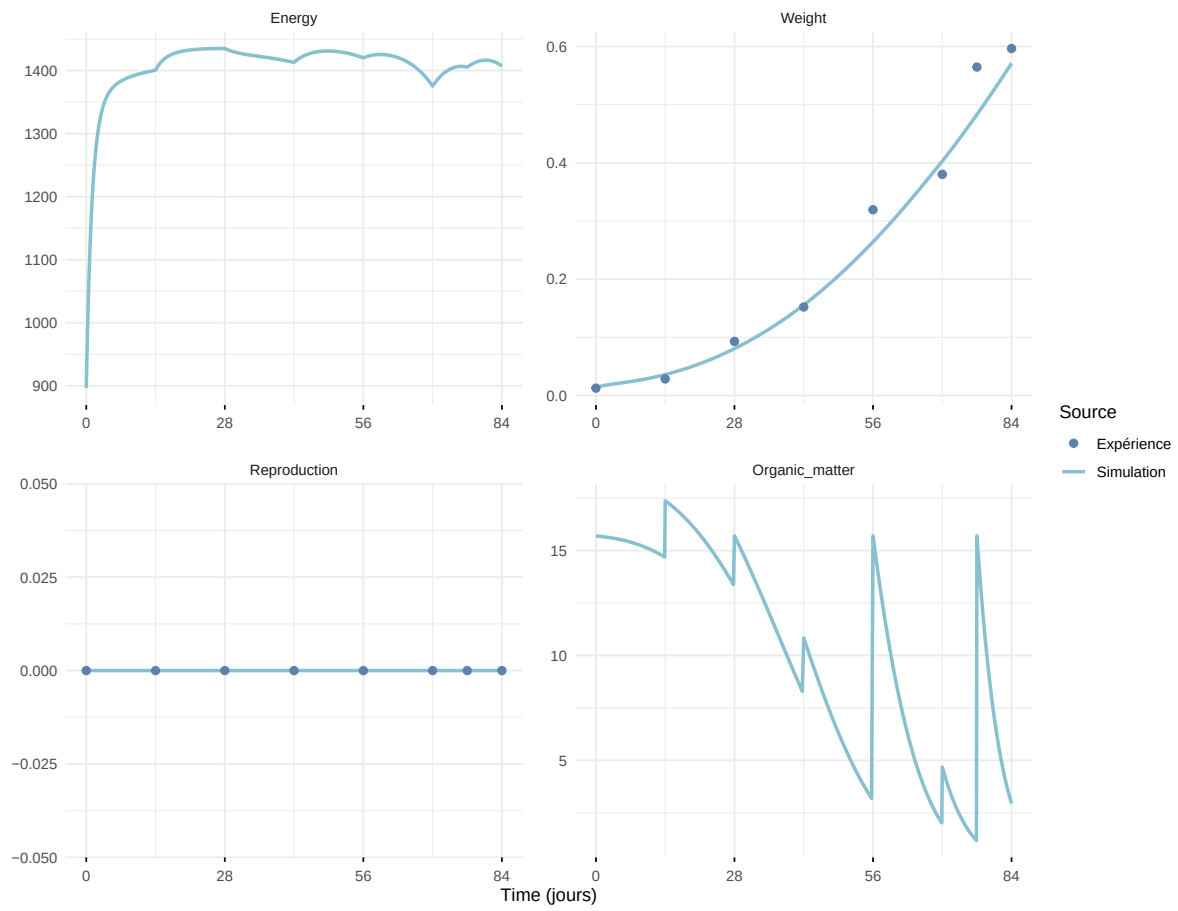
Expérience : Bart2019_XP1_1



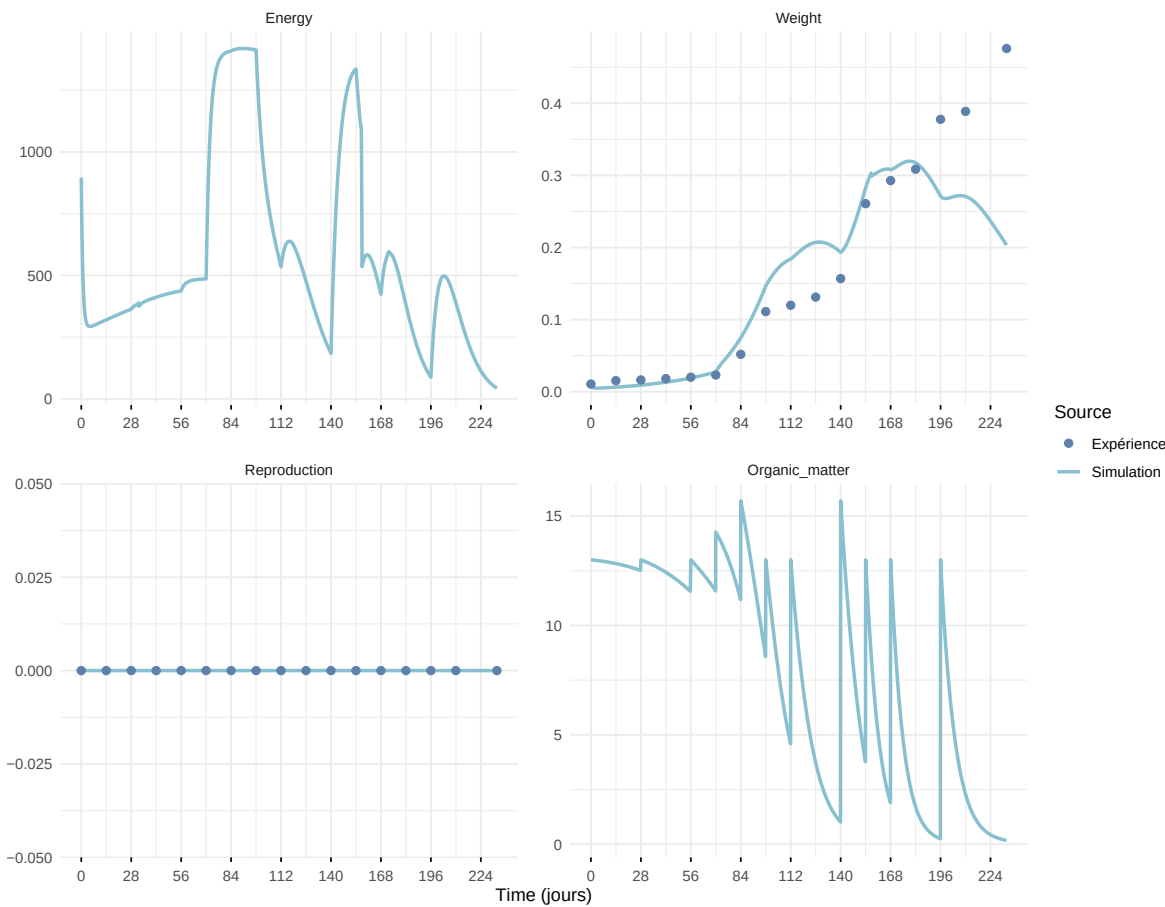
Expérience : Bart2019_XP1_2



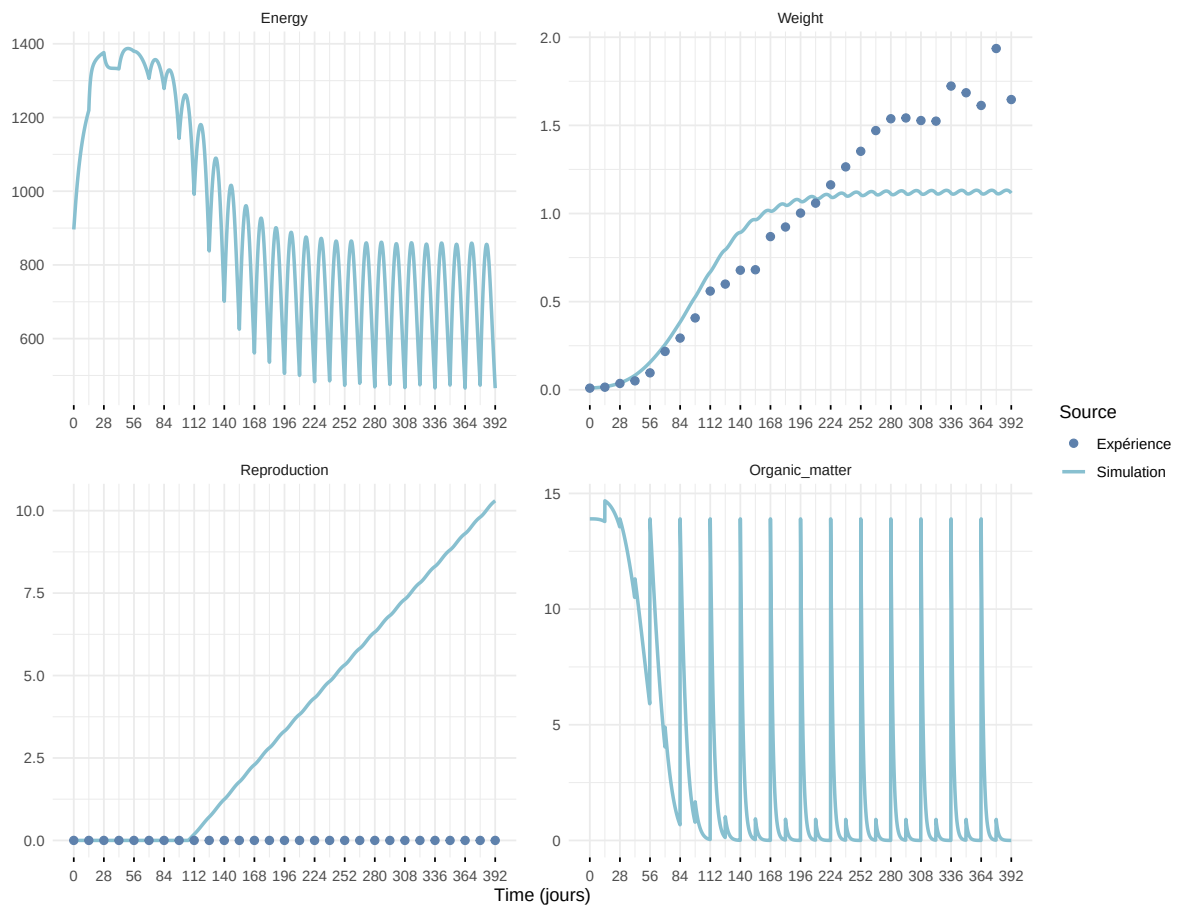
Expérience : Bart2019_XP2



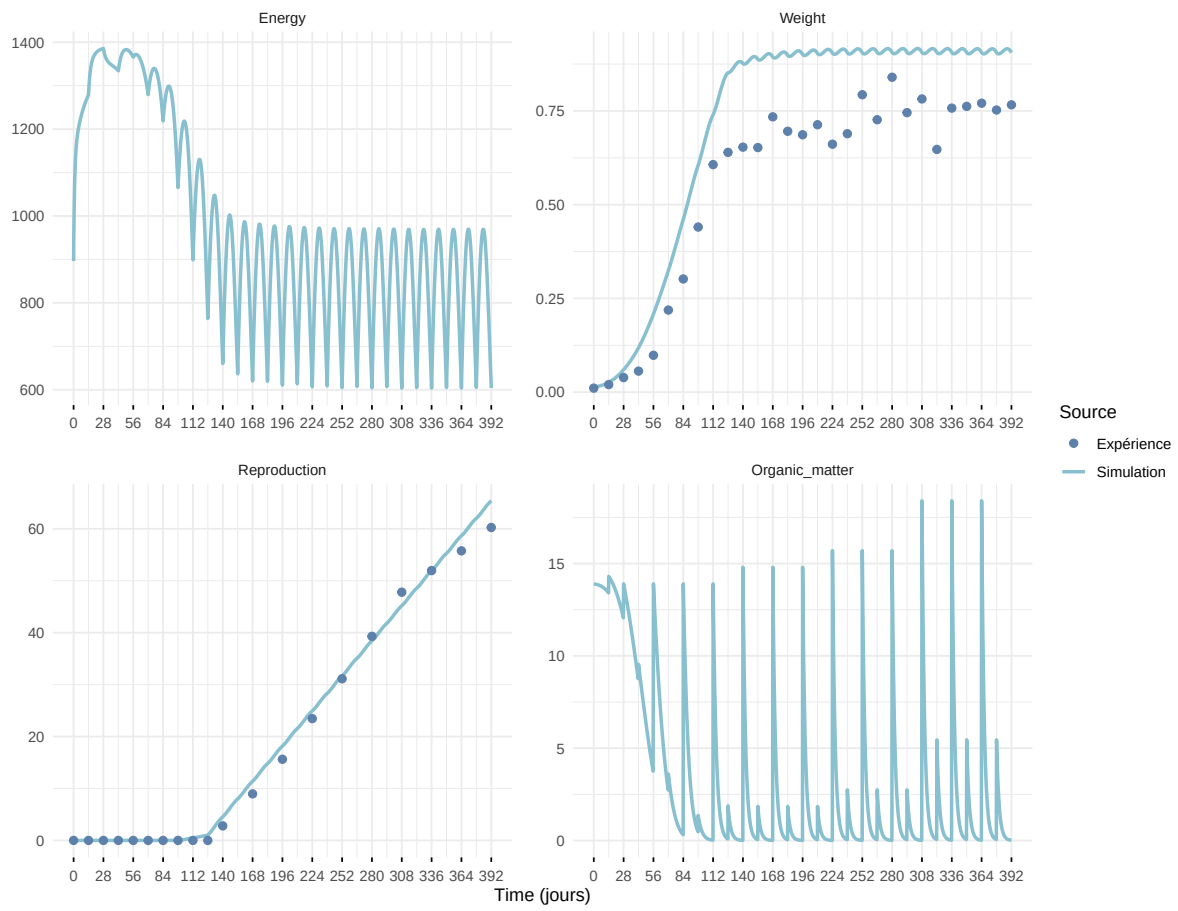
Expérience : Bart2019_XP3



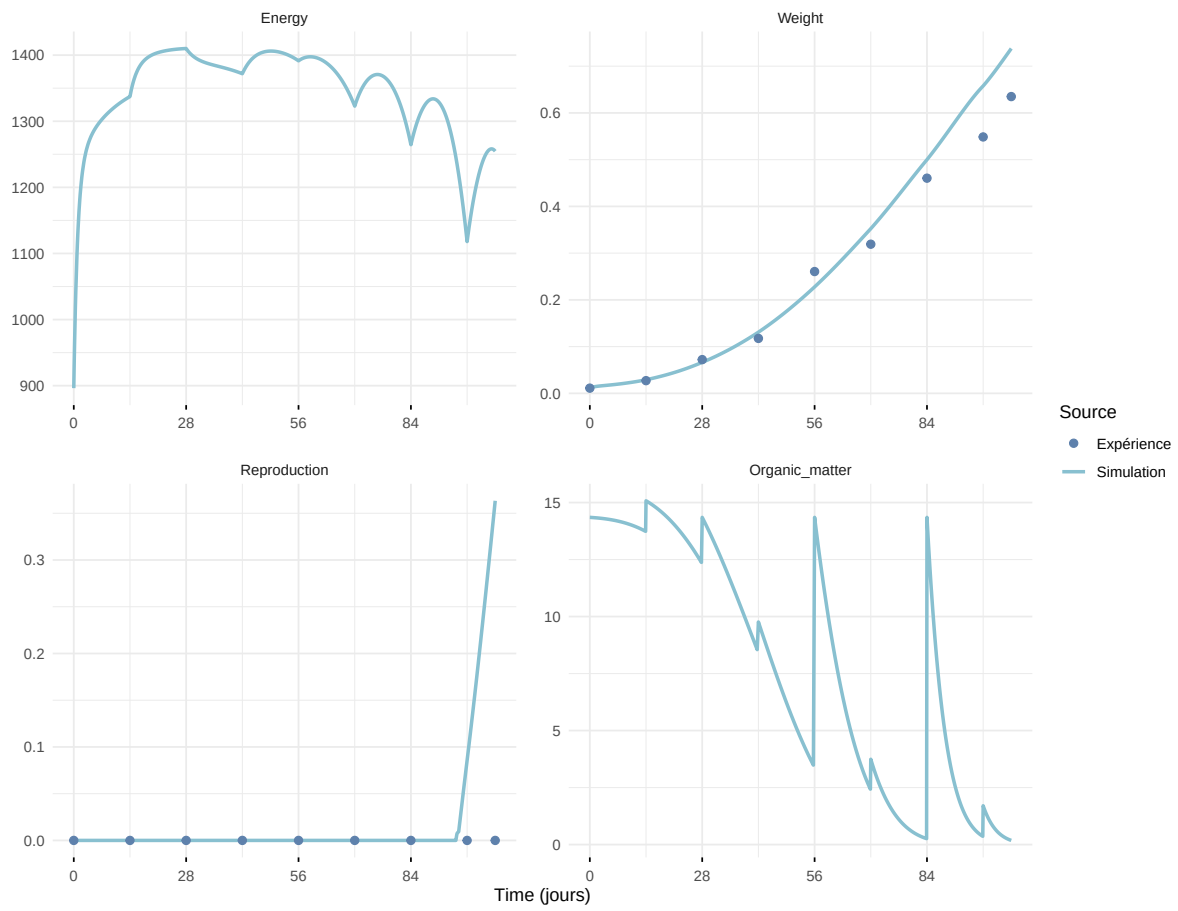
Expérience : Bart2019_XP4_1



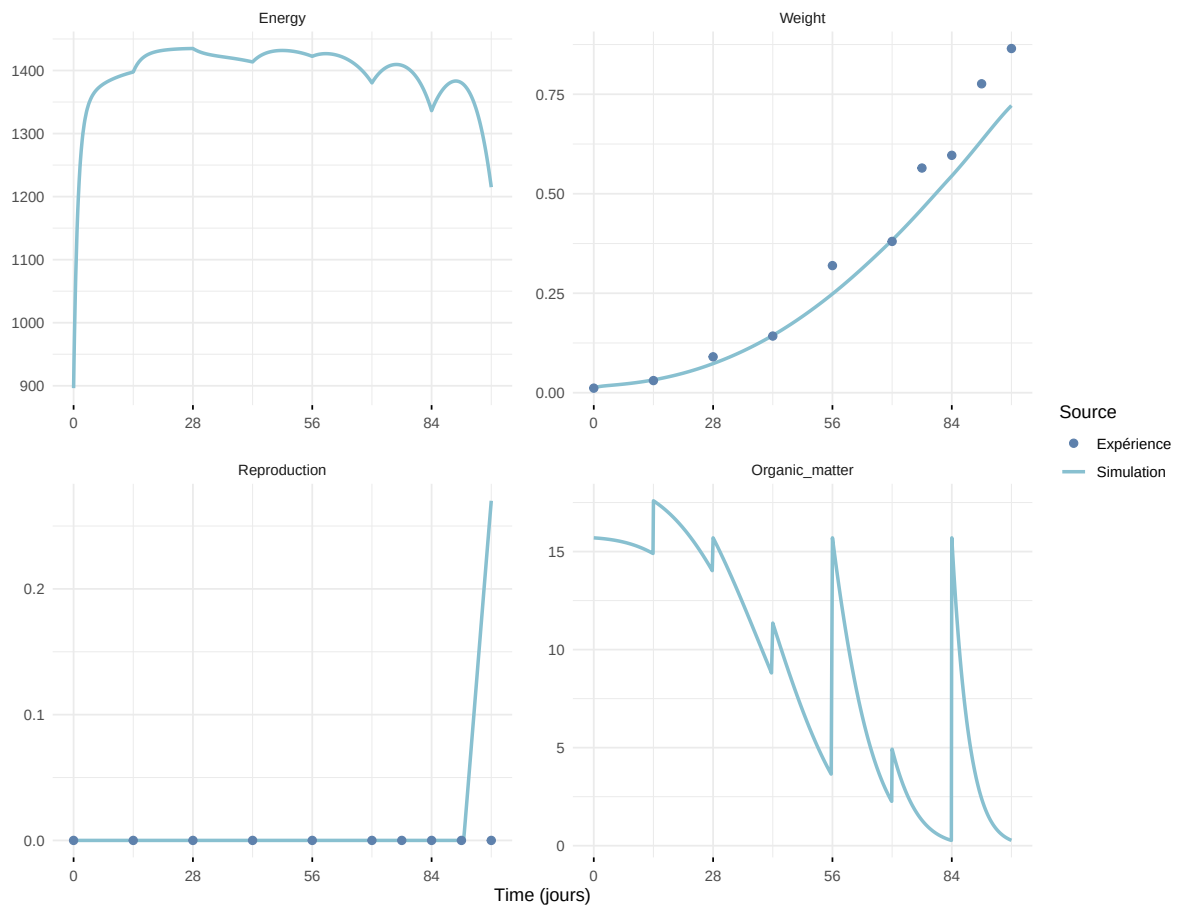
Expérience : Bart2019_XP4_2



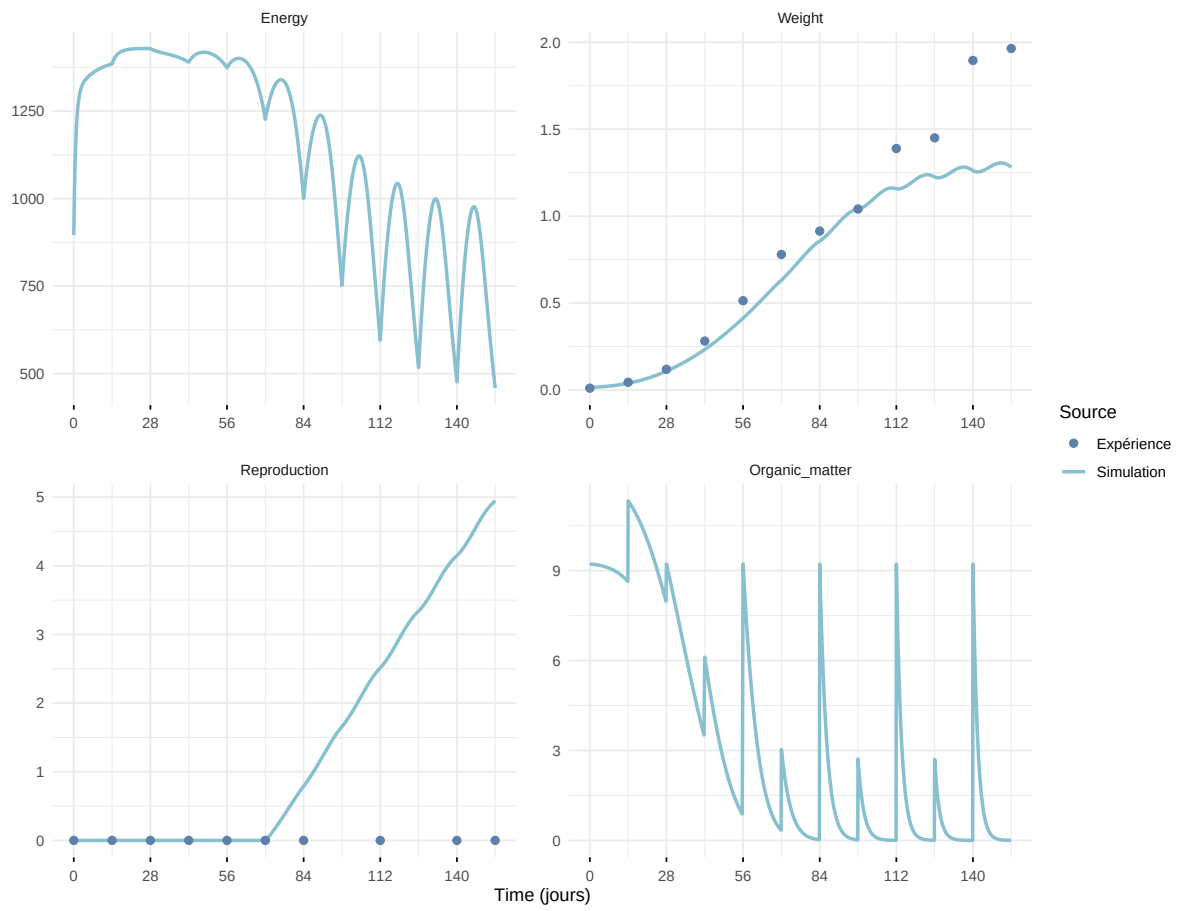
Expérience : Bart2019_XP5



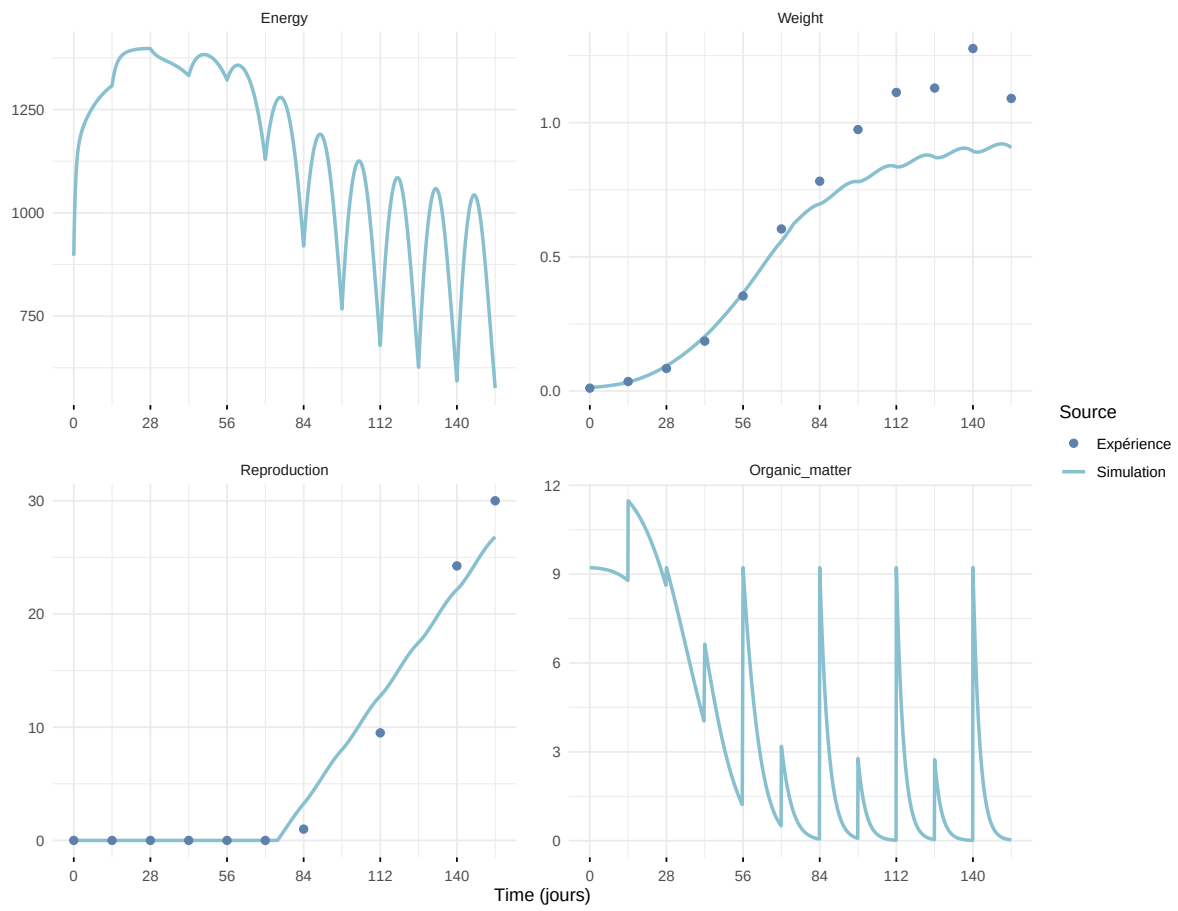
Expérience : Bart2020



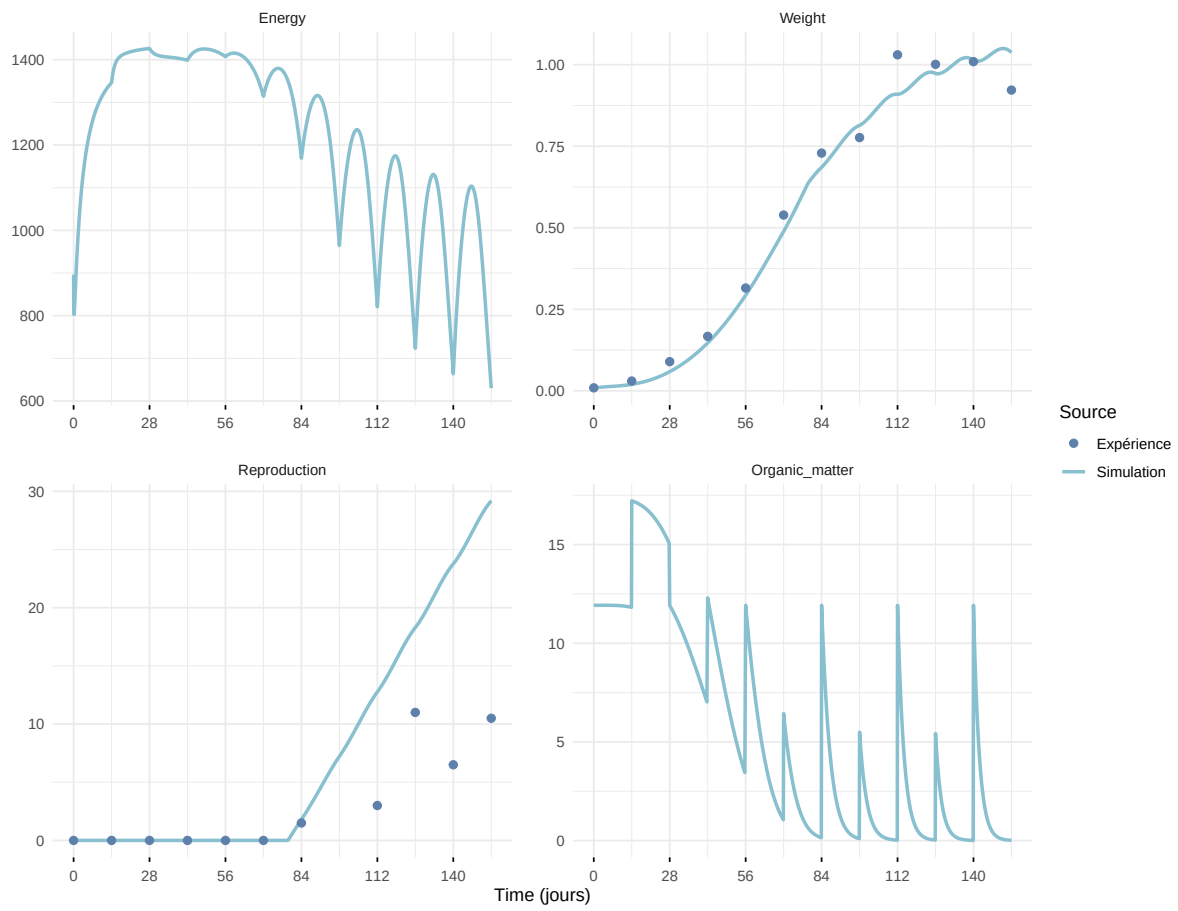
Expérience : Gollot2026_dens_D1



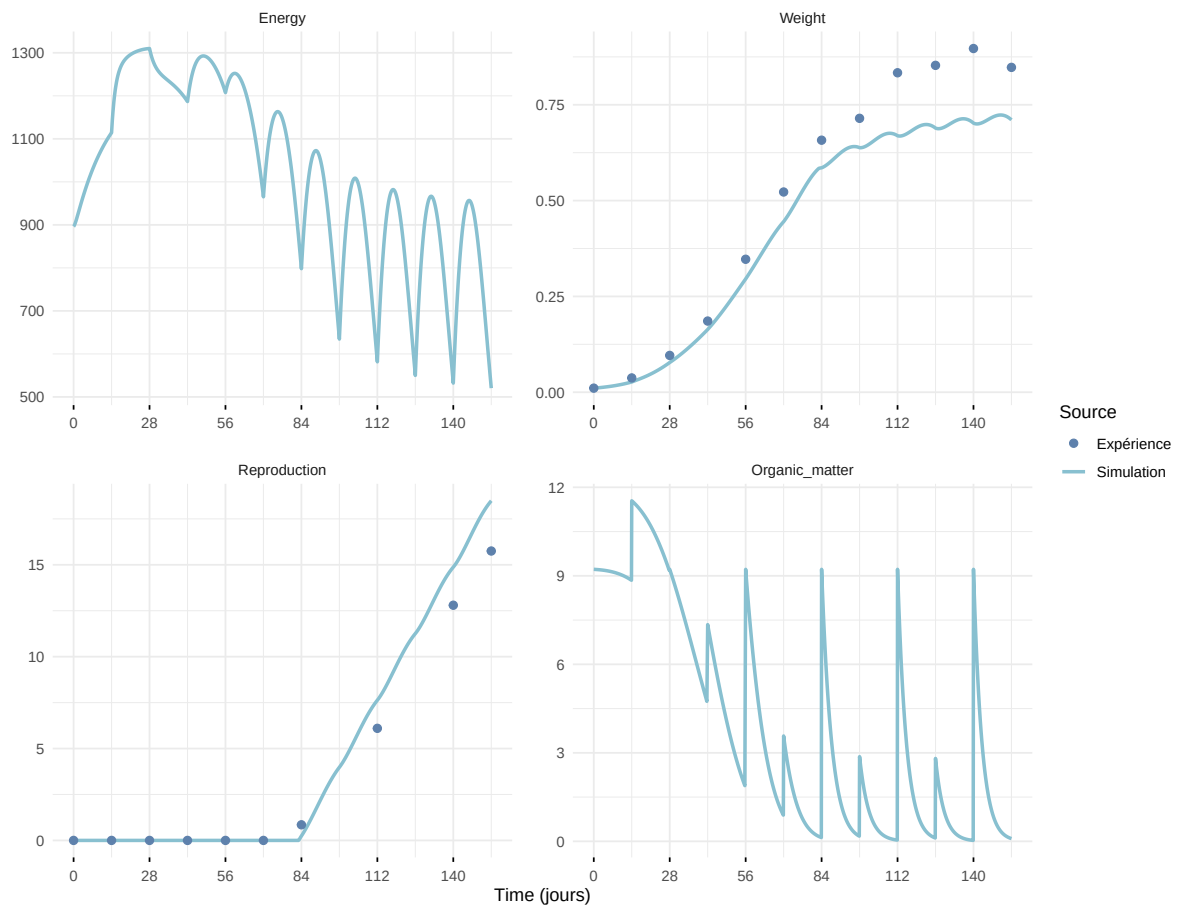
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

